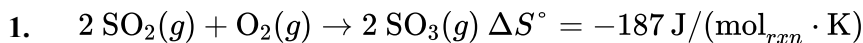


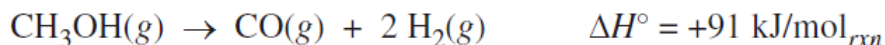
Chapter 08 Part 3: Entropy and Gibbs



Substance	Approximate S° (J/(mol · K))
$\text{SO}_2(g)$	?
$\text{O}_2(g)$	205
$\text{SO}_3(g)$	257

The reaction between SO_2 and O_2 is represented by the chemical equation above. The table provides the approximate absolute entropies, S° , for $\text{O}_2(g)$ and $\text{SO}_3(g)$. Which of the following mathematical expressions can be used to correctly calculate S° for $\text{SO}_2(g)$?

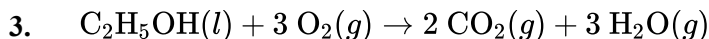
- (A) $S^\circ = -[-187 - 257 + 205] \text{ J}/(\text{mol} \cdot \text{K})$
- (B) $S^\circ = \frac{1}{2}[-187 + (2 \times 257) - 205] \text{ J}/(\text{mol} \cdot \text{K})$
- (C) $S^\circ = \frac{1}{2}[187 + (2 \times 257) - 205] \text{ J}/(\text{mol} \cdot \text{K})$ ✓
- (D) $S^\circ = [-187 + 257 - 205] \text{ J}/(\text{mol} \cdot \text{K})$



The reaction represented above goes essentially to completion. The reaction takes place in a rigid, insulated vessel that is initially at 600 K.

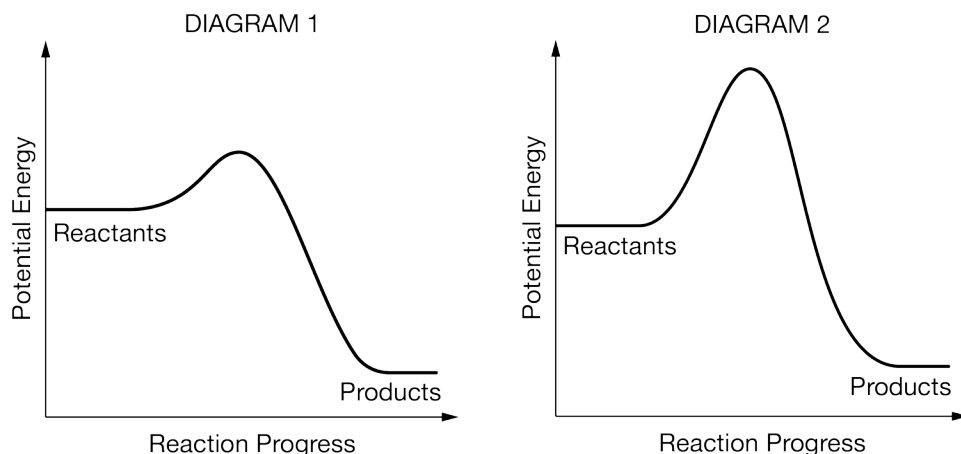
2. What can be inferred about ΔS° for the reaction at 600 K?
- (A) It must be positive, since the reaction is thermodynamically unfavorable at 600 K.
- (B) It must be negative, since there are more moles of products than reactants.
- (C) It must be positive, since ΔG° is negative and ΔH° is positive. ✓
- (D) It must be negative, since ΔG° is positive and ΔH° is positive.

Chapter 08 Part 3: Entropy and Gibbs



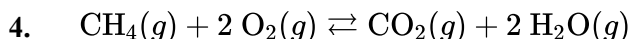
$$\Delta S^\circ = +217.7 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$$

$$\Delta H^\circ = -1235 \text{ kJ}/\text{mol}_{\text{rxn}}$$



The combustion of $\text{C}_2\text{H}_5\text{OH}$ is represented by the equation above and the standard entropy and enthalpy changes for the reaction are provided. When the reactants are combined at 25°C , essentially no $\text{CO}_2(g)$ or $\text{H}_2\text{O}(g)$ is produced after a few hours. Which of the diagrams above could best help explain the low yield of the reaction under these conditions, and why?

- (A) Diagram 1, because it represents a reaction that is not thermodynamically favorable with $\Delta G^\circ > 0$, regardless of its reaction rate.
- (B) Diagram 1, because it represents a reaction that reaches equilibrium quickly after a very small amount of the reactants is consumed.
- (C) Diagram 2, because it represents a reaction with a high activation energy barrier for molecules to overcome and a very slow reaction rate, even if it is thermodynamically favorable with $\Delta G^\circ < 0$. ✓
- (D) Diagram 2, because it represents a reaction that is thermodynamically favorable with $\Delta H^\circ < 0$, but the products formed are unstable and quickly revert to form reactants.



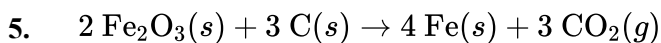
$$\Delta H^\circ_{\text{rxn}} = -803 \text{ kJ}/\text{mol}_{\text{rxn}}$$

$$\Delta S^\circ_{\text{rxn}} = -5 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$$

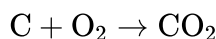
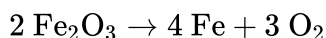
The chemical equation above represents the exothermic reaction of $\text{CH}_4(g)$ with $\text{O}_2(g)$. Which of the following best helps to explain why the reaction is thermodynamically favored ($\Delta G^\circ < 0$) at 2000 K and 1 atm?

Chapter 08 Part 3: Entropy and Gibbs

- (A) The total number of gaseous product molecules is less than the total number of gaseous reactant molecules, thus $\Delta S < 0$.
- (B) The total number of gaseous product molecules is greater than the total number of gaseous reactant molecules, thus $\Delta S > 0$.
- (C) The amount of energy released when the product bonds form is much less than the amount of energy needed to break the reactant bonds.
- (D) The amount of energy released when the product bonds form is much greater than the amount of energy needed to break the reactant bonds. ✓

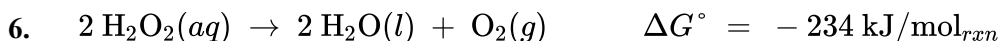


In a blast furnace, the reaction represented above occurs, producing $\text{Fe}(s)$ from its ore, $\text{Fe}_2\text{O}_3(s)$. The reaction is thermodynamically favorable and based on coupling the two reactions represented below.



Which of the following identifies a limitation in how the representations above describe a system of coupled reactions?

- (A) None of the equations are balanced.
- (B) The values of ΔS° for each reaction are not shown.
- (C) The values of ΔG° for each reaction are not shown. ✓
- (D) The coupled reactions lack a common intermediate species.



The value of ΔG° for the reaction represented above implies that the decomposition of $\text{H}_2\text{O}_2(aq)$ is thermodynamically favorable. However, $\text{H}_2\text{O}_2(aq)$ is typically stable for up to a year stored in a dark bottle at 298 K. The best explanation for this observation is that the decomposition reaction

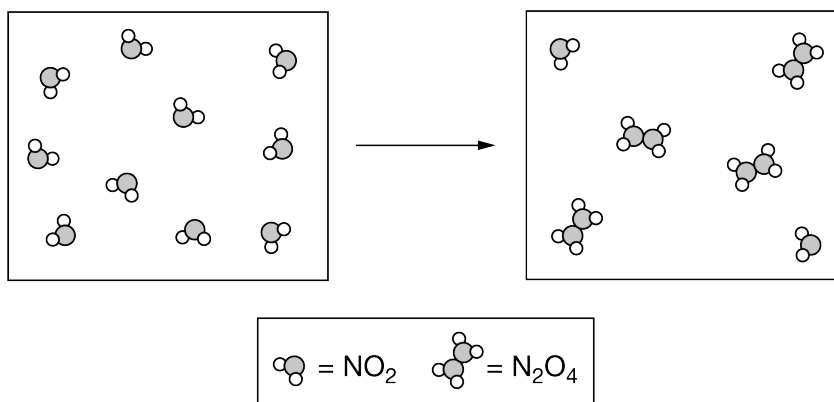
- (A) is only thermodynamically favorable in the presence of a catalyst
- (B) occurs with an increase in entropy because $\text{O}_2(g)$ is a product
- (C) is reversible and $\text{H}_2\text{O}_2(aq)$ is produced almost as fast as it decomposes
- (D) has a slow rate at 298 K because the activation energy is relatively high ✓

7. The equilibrium constant for a reaction is greater than 1.0 at temperatures above 500 K but less than 1.0 at temperatures below 500 K. What can be concluded about the values of ΔH° and ΔS° for the reaction? (Assume that ΔH° and ΔS° are independent of temperature.)

- (A) $\Delta H^\circ > 0$ and $\Delta S^\circ > 0$ ✓
- (B) $\Delta H^\circ > 0$ and $\Delta S^\circ < 0$
- (C) $\Delta H^\circ < 0$ and $\Delta S^\circ > 0$
- (D) $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$

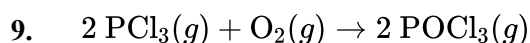
Chapter 08 Part 3: Entropy and Gibbs

8.



The diagram above represents the gas-phase reaction of $\text{NO}_2(g)$ to form $\text{N}_2\text{O}_4(g)$ at a certain temperature. Based on the diagram, which of the following best predicts and explains the sign of the entropy change for the reaction, $\Delta S^\circ_{\text{rxn}}$?

- (A) $\Delta S^\circ_{\text{rxn}}$ is negative because the number of N_2O_4 molecules increases as the reaction proceeds.
- (B) $\Delta S^\circ_{\text{rxn}}$ is negative because the number of molecules in the gas phase decreases as the reaction proceeds. ✓
- (C) $\Delta S^\circ_{\text{rxn}}$ is positive because the number of N_2O_4 molecules increases as the reaction proceeds.
- (D) $\Delta S^\circ_{\text{rxn}}$ is positive because the number of molecules in the gas phase decreases as the reaction proceeds.



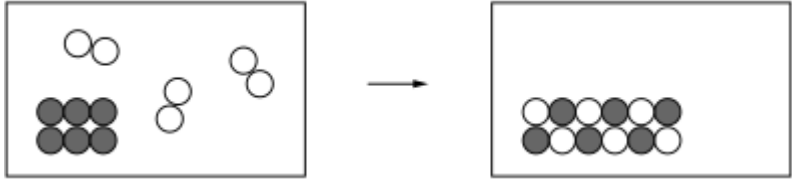
Substance	Approximate S° (J/(mol · K))
$\text{PCl}_3(g)$	310
$\text{O}_2(g)$	210
$\text{POCl}_3(g)$	330

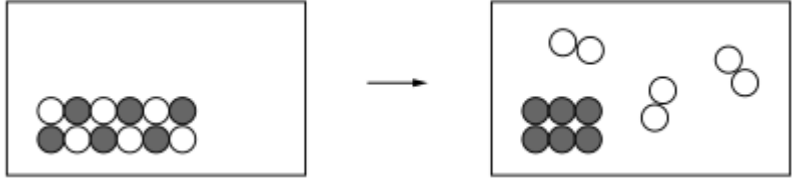
The oxidation of $\text{PCl}_3(g)$ is represented by the equation above, and the table provides the approximate values of the absolute molar entropies, S° , for these substances. Based on the information given, what is the approximate ΔS° for the reaction?

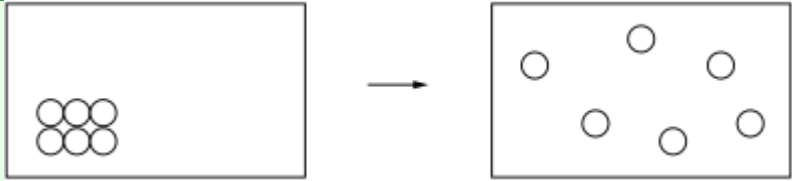
- (A) $+170 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$
- (B) $-170 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$ ✓
- (C) $+190 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$
- (D) $-190 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$

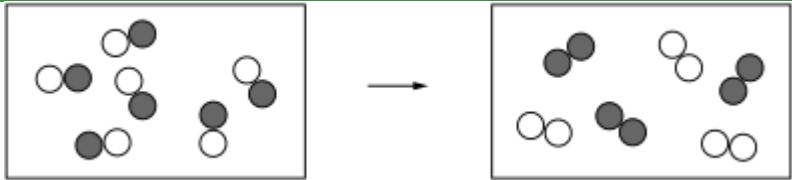
10. Which of the following pairs of diagrams represents the change with the greatest positive value of ΔS° ?

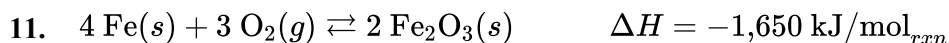
Chapter 08 Part 3: Entropy and Gibbs

(A) 

(B) 

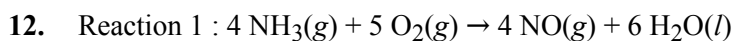
(C)  ✓

(D) 

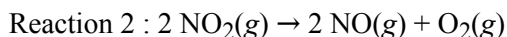


The oxidation of $\text{Fe}(s)$ is represented by the chemical equation above. Which of the following correctly explains whether or not the reaction is thermodynamically favorable?

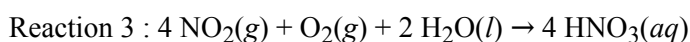
- (A) $\Delta S_{rxn} < 0$. Because ΔH is large and negative, the reaction will be thermodynamically favorable at low temperatures. ✓
- (B) $\Delta S_{rxn} < 0$. Because ΔH is large and negative, the reaction will not be thermodynamically favorable at any temperature.
- (C) $\Delta S_{rxn} > 0$. Because ΔH is large and negative, the reaction will be thermodynamically favorable at all temperatures.
- (D) $\Delta S_{rxn} > 0$. Because ΔH is large and negative, the reaction will not be thermodynamically favorable at any temperature.

Chapter 08 Part 3: Entropy and Gibbs

$$\Delta G^\circ = -1011 \text{ kJ/mol}_{rxn}$$

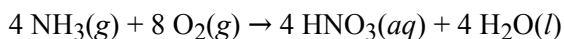


$$\Delta G^\circ = 71 \text{ kJ/mol}_{rxn}$$



$$\Delta G^\circ = -166 \text{ kJ/mol}_{rxn}$$

Based on the values of ΔG° for the three reactions represented above, what is the value of ΔG° for the reaction represented below?

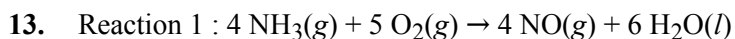


(A) $-1035 \text{ kJ/mol}_{rxn}$

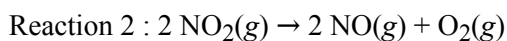
(B) $-1106 \text{ kJ/mol}_{rxn}$

(C) $-1248 \text{ kJ/mol}_{rxn}$

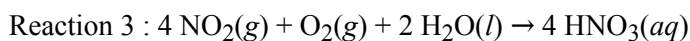
(D) $-1319 \text{ kJ/mol}_{rxn}$



$$\Delta G^\circ = -1010 \text{ kJ/mol}_{rxn}$$

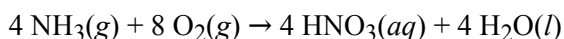


$$\Delta G^\circ = 70 \text{ kJ/mol}_{rxn}$$

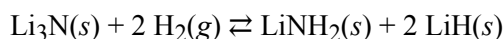


$$\Delta G^\circ = -170 \text{ kJ/mol}_{rxn}$$

Based on the values of ΔG° for the three reactions represented above, what is the value of ΔG° for the reaction represented below?



Chapter 08 Part 3: Entropy and Gibbs

(A) $-1040 \text{ kJ/mol}_{rxn}$ (B) $-1110 \text{ kJ/mol}_{rxn}$ (C) $-1250 \text{ kJ/mol}_{rxn}$ (D) $-1320 \text{ kJ/mol}_{rxn}$ ✓

$$\Delta H^\circ = -192 \text{ kJ/mol}_{rxn}$$

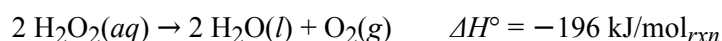
Because pure H_2 is a hazardous substance, safer and more cost effective techniques to store it as a solid for shipping purposes have been developed. One such method is the reaction represented above, which occurs at 200°C .

14. Which of the following is the most likely reason that the reaction occurs at a significant rate only if the temperature of the reaction mixture is greater than 200°C ?

(A) The reaction is exothermic.

(B) ΔS° for the reaction is negative.

(C) The reaction has a high activation energy. ✓

(D) $\Delta G^\circ < 0$ when $T < 200^\circ\text{C}$.

The decomposition of $\text{H}_2\text{O}_2(aq)$ is represented by the equation above. A student monitored the decomposition of a 1.0 L sample of $\text{H}_2\text{O}_2(aq)$ at a constant temperature of 300. K and recorded the concentration of H_2O_2 as a function of time. The results are given in the table below.

Time (s)	$[\text{H}_2\text{O}_2]$
0	2.7
200.	2.1
400.	1.7
600.	1.4

15. The reaction is thermodynamically favorable. The signs of ΔG° and ΔS° for the reaction are which of the following?

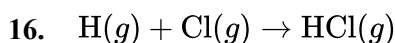
Chapter 08 Part 3: Entropy and Gibbs

(A)	ΔG°	ΔS°
	Positive	Positive

(B)	ΔG°	ΔG°	✓
	Negative	Positive	

(C)	ΔG°	ΔS°
	Positive	Negative

(D)	ΔG°	ΔS°
	Negative	Negative



The formation of $\text{HCl}(g)$ from its atoms is represented by the equation above. Which of the following best explains why the reaction is thermodynamically favored?

- (A) $\Delta G > 0$ because energy is released as the bond between the H and Cl atoms forms, and entropy increases because the number of gaseous product particles is less than the number of gaseous reactant particles.
- (B) $\Delta G > 0$ because energy is absorbed as the bond between the H and Cl atoms forms, and entropy decreases because the number of gaseous product particles is less than the number of gaseous reactant particles.
- (C) $\Delta G < 0$ because although energy is absorbed as the bond between the H and Cl atoms forms, entropy increases because the number of gaseous product particles is less than the number of gaseous reactant particles.

- (D) $\Delta G < 0$ because although entropy decreases because the number of gaseous product particles is less than the number of gaseous reactant particles, energy is released as the bond between the H and Cl atoms forms. ✓

Chapter 08 Part 3: Entropy and Gibbs

17. A cube of ice is added to some hot water in a rigid, insulated container, which is then sealed. There is no heat exchange with the surroundings. What has happened to the total energy and the total entropy when the system reaches equilibrium?

(A)	<u>Energy</u>	<u>Entropy</u>
	Remains constant	Remains constant

(B)	<u>Energy</u>	<u>Entropy</u>
	Remains constant	Decreases

(C)	<u>Energy</u>	<u>Entropy</u>	✓
	Remains constant	Increases	

(D)	<u>Energy</u>	<u>Entropy</u>
	Decreases	Increases

(E)	<u>Energy</u>	<u>Entropy</u>
	Increases	Decreases

Chapter 08 Part 3: Entropy and Gibbs



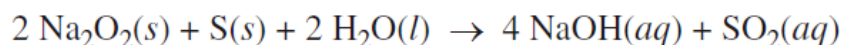
The elements K and Cl react directly to form the compound KCl according to the equation above. Refer to the information above and the table below to answer the questions that follow.

Process	ΔH° (kJ/mol _{rxn})
$\text{K}(s) \rightarrow \text{K}(g)$	v
$\text{K}(g) \rightarrow \text{K}^+(g) + e^-$	w
$\text{Cl}_2(g) \rightarrow 2 \text{Cl}(g)$	x
$\text{Cl}(g) + e^- \rightarrow \text{Cl}^-(g)$	y
$\text{K}^+(g) + \text{Cl}^-(g) \rightarrow \text{KCl}(s)$	z

18. It is observed that the reaction producing KCl from its elements goes essentially to completion. Which of the following is a true statement about the thermodynamic favorability of the reaction?

- (A) The reaction is favorable and driven by an enthalpy change only. ✓
- (B) The reaction is unfavorable and driven by an entropy change only.
- (C) The reaction is favorable and driven by both enthalpy and entropy changes.
- (D) The reaction is unfavorable due to both enthalpy and entropy changes.

When water is added to a mixture of $\text{Na}_2\text{O}_2(s)$ and $\text{S}(s)$, a redox reaction occurs, as represented by the equation below.

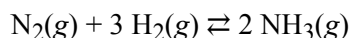


$$\Delta H_{298}^\circ = -610 \text{ kJ/mol}_{\text{rxn}} ; \Delta S_{298}^\circ = -7.3 \text{ J/(K} \cdot \text{mol}_{\text{rxn}})$$

19. Which of the following statements about the thermodynamic favorability of the reaction at 298 K is correct?

Chapter 08 Part 3: Entropy and Gibbs

- (A) It is thermodynamically unfavorable.
- (B) It is thermodynamically favorable and is driven by ΔS° only.
- (C) It is thermodynamically favorable and is driven by ΔH° only. ✓
- (D) It is thermodynamically favorable and is driven by both ΔH° and ΔS° .
-

20.

$$\Delta H_{298}^\circ = -92 \text{ kJ/mol}_{\text{rxn}}; \Delta G_{298}^\circ = -33 \text{ kJ/mol}_{\text{rxn}}$$

Consider the reaction represented above at 298 K. When equal volumes of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$, each at 1 atm, are mixed in a closed container at 298 K, no formation of $\text{NH}_3(\text{g})$ is observed. Which of the following best explains the observation?

- (A) The $\text{N}_2(\text{g})$ and the $\text{H}_2(\text{g})$ must be mixed in a 1:3 ratio for a reaction to occur.
- (B) A high activation energy makes the forward reaction extremely slow at 298 K. ✓
- (C) The reaction has an extremely small equilibrium constant, thus almost no product will form.
- (D) The reverse reaction has a lower activation energy than the forward reaction, so the forward reaction does not occur.

- 21.** When solid ammonium chloride, $\text{NH}_4\text{Cl}(\text{s})$, is added to water at 25°C , it dissolves and the temperature of the solution decreases. Which of the following is true for the values of ΔH and ΔS for the dissolving process?

Chapter 08 Part 3: Entropy and Gibbs

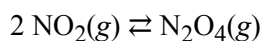
(A)	ΔH	ΔS	✓
	Positive	Positive	

(B)	ΔH	ΔS
	Positive	Negative

(C)	ΔH	ΔS
	Positive	Equal to zero

(D)	ΔH	ΔS
	Negative	Positive

(E)	ΔH	ΔS
	Negative	Negative



dark brown colorless

The dimerization of $\text{NO}_2(g)$, an exothermic process, is represented by the equation above.

22. The forward reaction is thermodynamically favored at which of the following temperatures?

(A) All temperatures

(B) Low temperatures only

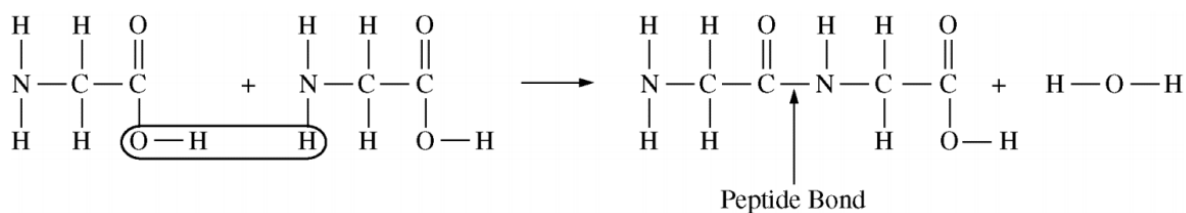
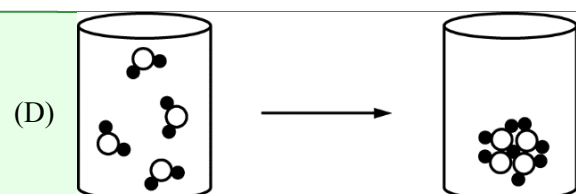
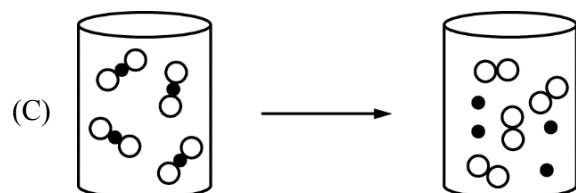
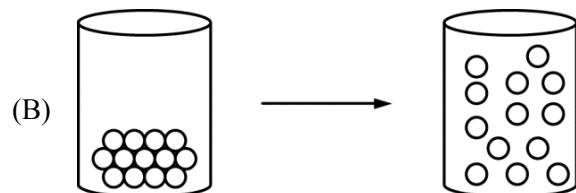
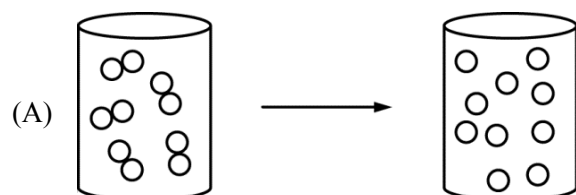
(C) High temperatures only

(D) No temperature



Chapter 08 Part 3: Entropy and Gibbs

23. Which of the following particulate representations shows a process during which the entropy of the system decreases?



Two molecules of the amino acid glycine join through the formation of a peptide bond, as shown above. The thermodynamic data for the reaction are listed in the following table.

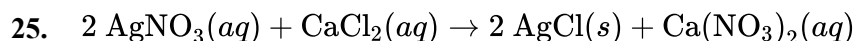
ΔG_{298}°	ΔH_{298}°	ΔS_{298}°
+15 kJ/mol _{rxn}	+12 kJ/mol _{rxn}	-10 J/(K·mol _{rxn})

24. Under which of the following temperature conditions is the reaction thermodynamically favored?

Chapter 08 Part 3: Entropy and Gibbs

- (A) It is only favored at high temperatures.
(B) It is only favored at low temperatures.
(C) It is favored at all temperatures.

(D) It is not favored at any temperature. ✓



Substance	Approximate S° (J/(mol · K))
$\text{AgNO}_3(aq)$	220
$\text{CaCl}_2(aq)$	60.
$\text{AgCl}(s)$	96
$\text{Ca}(\text{NO}_3)_2(aq)$	240

The reaction between AgNO_3 and CaCl_2 is represented by the equation above, and the table provides the approximate S° values for the reactants and products. Which of the following is the approximate ΔS° for the reaction?

(A) $-68 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$ ✓

(B) $+68 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$

(C) $-56 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$

(D) $+56 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$

26. In which of the following processes will ΔS° be negative?

(A) $\text{C}_2\text{H}_5\text{OH}(l) \rightarrow \text{C}_2\text{H}_5\text{OH}(g)$

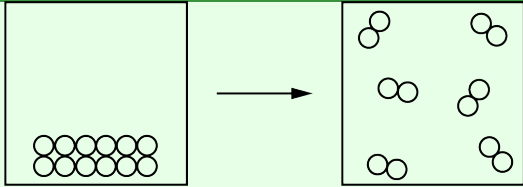
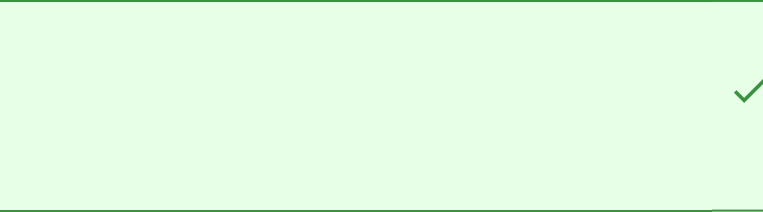
(B) $\text{NaCl}(s) \rightarrow \text{NaCl}(l)$

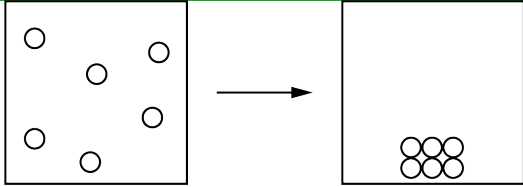
(C) $\text{CO}_2(s) \rightarrow \text{CO}_2(g)$

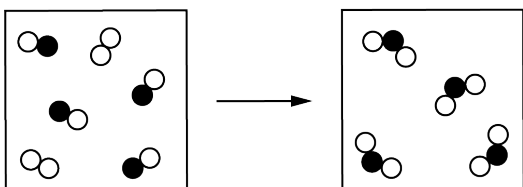
(D) $\text{Cl}_2(g) \rightarrow \text{Cl}_2(l)$ ✓

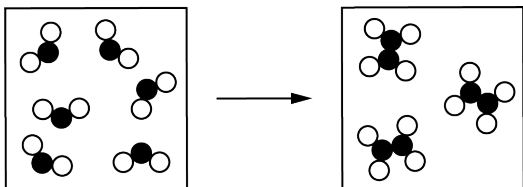
27. Which of the following particle diagrams represents a process during which the entropy of the system increases?

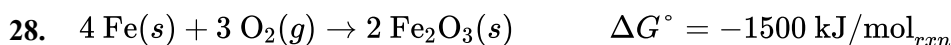
Chapter 08 Part 3: Entropy and Gibbs

(A)  

(B) 

(C) 

(D) 



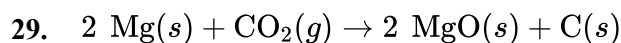
The reaction of iron with oxygen to form rust is represented by the equation shown above. A student cleans two iron nails and places each nail in a capped test tube. The following table gives the experimental conditions and the student's observations after one week at room temperature.

Test Tube	Experimental Conditions Inside the Capped Test Tube	Observations
1	Dry air	No visible rust on nail
2	Air and water	Rust suspended in the water and on the nail

The student claims that the formation of rust in test tube 2 shows that the reaction is thermodynamically favored. Which of the following justifications should the student use to explain why rust did not form in test tube 1?

- (A) The reaction does not occur at an observable rate when water is not present because it proceeds through a mechanism with a high activation energy. 
- (B) The reaction is less thermodynamically favored because the Gibbs free energy of the product is greater when water is not present.
- (C) The product is not formed in measurable quantities because the equilibrium constant for the reaction when water is not present is significantly less than one.
- (D) The rate of reaction is much slower because the oxygen molecules collide with the iron surface with less energy when water is not present.

Chapter 08 Part 3: Entropy and Gibbs



$$\Delta H_{rxn}^{\circ} < 0; \quad \Delta G_{rxn}^{\circ} < 0 \text{ at } 298\text{K}$$

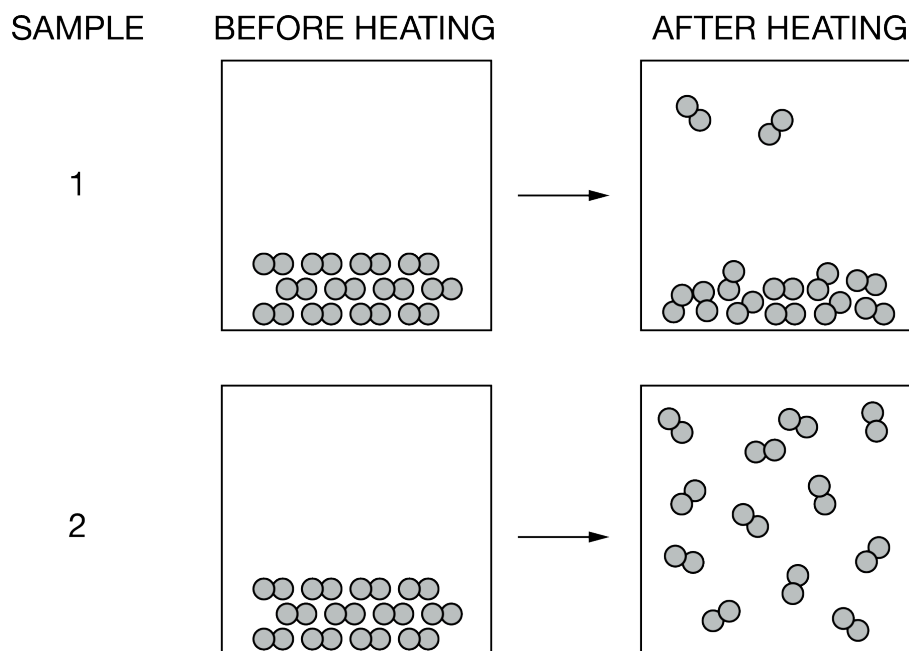
Which of the following is true about the reaction represented above?

- (A) At 298 K, it is driven by enthalpy only. ✓
- (B) At 298 K, it is driven by entropy only.
- (C) At 298 K, it is driven by both enthalpy and entropy.
- (D) It is thermodynamically favorable at all temperatures.

30. Which of the following equations represents a reaction with a negative ΔS° ?

- (A) $2 \text{NO}(g) + \text{O}_2(g) \rightarrow 2 \text{NO}_2(g)$ ✓
- (B) $2 \text{KClO}_3(s) \rightarrow 2 \text{KCl}(s) + 3 \text{O}_2(g)$
- (C) $\text{PCl}_5(g) \rightarrow \text{PCl}_3(g) + \text{Cl}_2(g)$
- (D) $\text{C}_3\text{H}_8(g) + 5 \text{O}_2(g) \rightarrow 3 \text{CO}_2(g) + 4 \text{H}_2\text{O}(g)$

31.

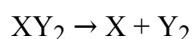


The particle diagrams above represent a change in physical state that occurred after heating two separate solid samples of a diatomic element. Which of the following best compares the relative magnitude of ΔS° and gives the sign for the entropy change undergone by each sample, and why?

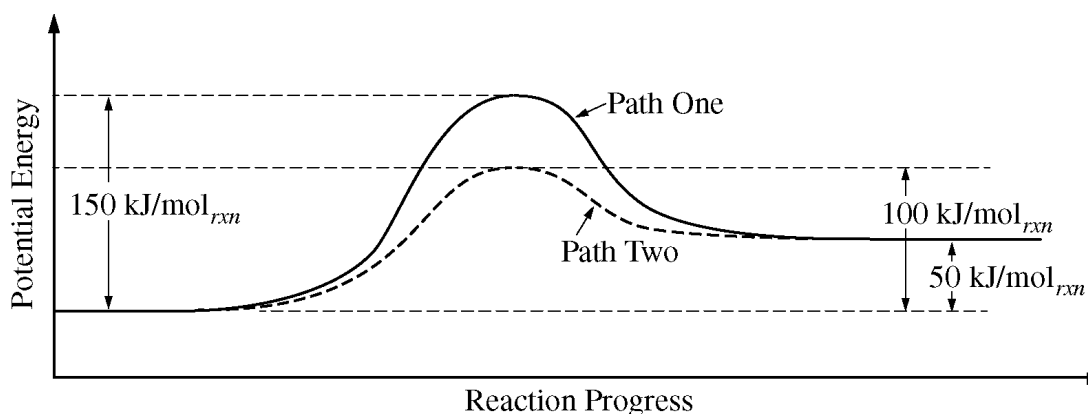
Chapter 08 Part 3: Entropy and Gibbs

- (A) The entropy values for both samples are positive and $\Delta S^\circ_{\text{sample 2}} = \Delta S^\circ_{\text{sample 1}}$, because the changes observed for both samples required a net input of energy from the surroundings.
- (B) The entropy values for both samples are negative and $\Delta S^\circ_{\text{sample 2}} = \Delta S^\circ_{\text{sample 1}}$, because the changes observed for both samples released energy to the surroundings.
- (C) The entropy values for both samples are positive and $\Delta S^\circ_{\text{sample 2}} > \Delta S^\circ_{\text{sample 1}}$, because the change in the spatial distribution of the molecules was greater for sample 2 than for sample 1. ✓
- (D) The entropy values for both samples are negative and $\Delta S^\circ_{\text{sample 2}} < \Delta S^\circ_{\text{sample 1}}$, because the change in the spatial distribution of the molecules was greater for sample 2 than for sample 1.

The following questions relate to the below information.



The equation above represents the decomposition of a compound XY_2 . The diagram below shows two reaction profiles (path one and path two) for the decomposition of XY_2 .



32. The reaction is thermodynamically favorable under standard conditions at 298 K. Therefore, the value of ΔS° for the reaction must be
- (A) equal to zero
- (B) equal to $\Delta H^\circ/298 \text{ K}$
- (C) greater than $\Delta H^\circ/298 \text{ K}$ ✓
- (D) less than $\Delta H^\circ/298 \text{ K}$
33. Which of the following reactions is not thermodynamically favored at low temperatures but becomes favored as the temperature increases?

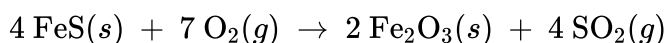
Chapter 08 Part 3: Entropy and Gibbs

(A)	Reaction	ΔH° (kJ/mol _{rxn})	ΔS° (J/(mol _{rxn} K))
	$2 \text{ CO(g)} + \text{O}_2\text{(g)} \rightarrow 2 \text{ CO}_2\text{(g)}$	-566	-173

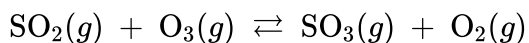
(B)	Reaction	ΔH° (kJ/mol _{rxn})	ΔS° (J/(mol _{rxn} K))
	$2 \text{ H}_2\text{O(g)} \rightarrow 2 \text{ H}_2\text{(g)} + \text{O}_2\text{(g)}$	484	90.0

(C)	Reaction	ΔH° (kJ/mol _{rxn})	ΔS° (J/(mol _{rxn} K))
	$2 \text{ N}_2\text{O(g)} \rightarrow 2 \text{ N}_2\text{(g)} + \text{O}_2\text{(g)}$	-164	149

(D)	Reaction	ΔH° (kJ/mol _{rxn})	ΔS° (J/(mol _{rxn} K))
	$\text{PbCl}_2\text{(s)} \rightarrow \text{Pb}^{2+}\text{(aq)} + 2 \text{ Cl}^-\text{(aq)}$	23.4	-12.5



Due to the presence of FeS(s) as an impurity, the combustion of some types of coal results in the formation of SO₂(g), as represented by the equation above. Also, SO₂(g) can react with O₃(g) to form SO₃(g), as represented by the equation below.



$$\Delta H_{298}^\circ = -242 \text{ kJ/mol}_{\text{rxn}} ; \Delta S_{298}^\circ = -25 \text{ J/(K} \cdot \text{mol}_{\text{rxn}})$$

34. Which of the following is most likely to be true about the reaction between SO₂(g) and O₃(g) at 298 K?

(A) $\Delta G^\circ < 0$ and $T\Delta S^\circ$ is smaller in magnitude than ΔH° .

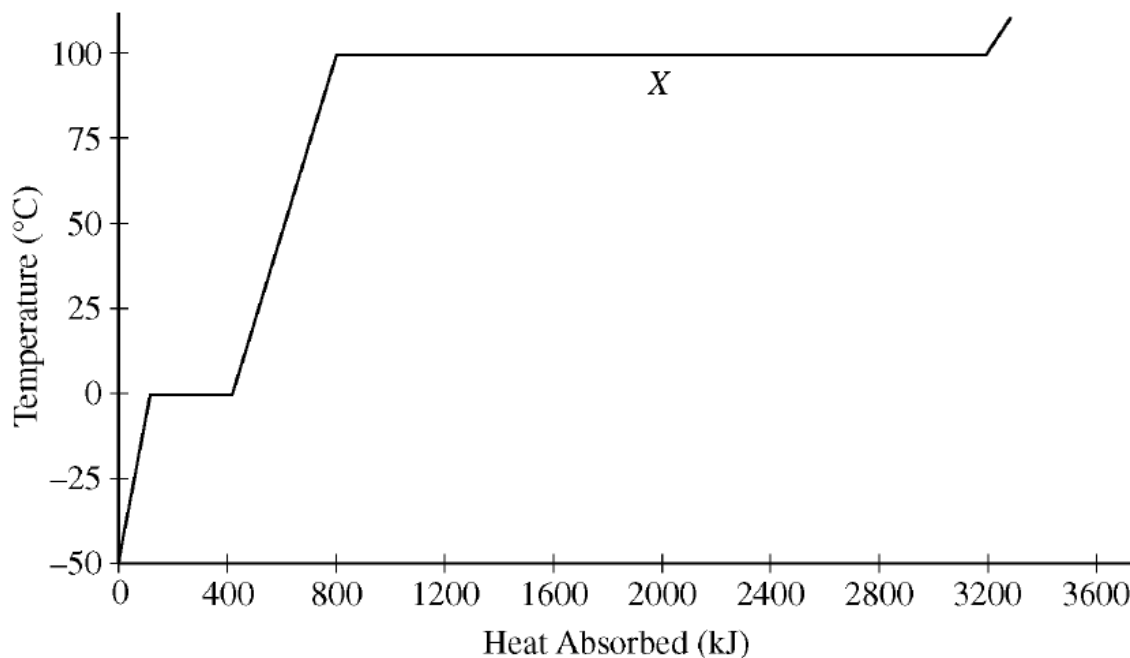
(B) $\Delta G^\circ < 0$ and $T\Delta S^\circ$ is larger in magnitude than ΔH° .

(C) $\Delta G^\circ > 0$ and $T\Delta S^\circ$ is smaller in magnitude than ΔH° .

(D) $\Delta G^\circ > 0$ and $T\Delta S^\circ$ is larger in magnitude than ΔH° .

Chapter 08 Part 3: Entropy and Gibbs

35.



At 1.0 atm a sample of ice is heated to liquid water and then to water vapor. The heating curve is shown in the graph above. Which of the following lists the signs for the changes in enthalpy and entropy for the process corresponding to segment *X*, going from left to right?

Chapter 08 Part 3: Entropy and Gibbs

(A)	ΔH°	ΔS°
	Positive	Negative

(B)	ΔH°	ΔS°	✓
	Positive	Positive	

(C)	ΔH°	ΔS°
	Negative	Negative

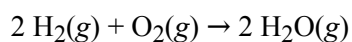
(D)	ΔH°	ΔS°
	Negative	Positive

36. A student observes that the equilibrium constant for a reaction is greater than 1.0 at temperatures below 500 K but less than 1.0 at temperatures above 500 K. What can the student conclude about the values of ΔH° and ΔS° for the reaction? (Assume that ΔH° and ΔS° are independent of temperature.)

- (A) $\Delta H^\circ > 0$ and $\Delta S^\circ > 0$
 (B) $\Delta H^\circ > 0$ and $\Delta S^\circ < 0$
 (C) $\Delta H^\circ < 0$ and $\Delta S^\circ > 0$

- (D) $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$ ✓

37.



For the reaction represented above at 25°C, what are the signs of ΔH° , ΔS° , and ΔG° ?

Chapter 08 Part 3: Entropy and Gibbs

(A)	ΔH°	ΔS°	ΔG°
	+	+	+

(B)	ΔH°	ΔS°	ΔG°
	+	+	-

(C)	ΔH°	ΔS°	ΔG°
	+	-	-

(D)	ΔH°	ΔS°	ΔG°	✓
	-	-	-	

(E)	ΔH°	ΔS°	ΔG°
	-	-	+

38. Quantity that would be zero for a pure, perfect crystal at 0 K

- (A) Activation energy
- (B) Enthalpy of formation
- (C) Entropy
- (D) Gibbs free energy
- (E) Lattice energy

39. Which of the following equations represents a reaction for which the standard entropy change is positive ($\Delta S^\circ > 0$)?

Chapter 08 Part 3: Entropy and Gibbs

- (A) $3 \text{O}_2(g) \rightarrow 2 \text{O}_3(g)$
(B) $2 \text{H}_2(g) + \text{O}_2(g) \rightarrow 2 \text{H}_2\text{O}(l)$
(C) $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$ ✓
(D) $\text{I}_2(g) + 2 \text{K}(s) \rightarrow 2 \text{KI}(s)$

40. For which of the following processes would ΔS have a negative value?

- I. $2 \text{Fe}_2\text{O}_3(s) \rightarrow 4 \text{Fe}(s) + 3 \text{O}_2(g)$
II. $\text{Mg}^{2+} + 2 \text{OH}^- \rightarrow \text{Mg}(\text{OH})_2(s)$
III. $\text{H}_2(g) + \text{C}_2\text{H}_4(g) \rightarrow \text{C}_2\text{H}_6(g)$
(A) I only
(B) I and II only
(C) I and III only
(D) II and III only ✓
(E) I, II, and III

41. Which of the following processes involves the greatest increase in entropy?

- (A) $\text{SO}_3(g) + \text{H}_2(g) \rightarrow \text{SO}_2(g) + \text{H}_2\text{O}(g)$
(B) $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$
(C) $\text{Ag}^+(aq) + \text{Cl}^-(aq) \rightarrow \text{AgCl}(s)$
(D) $\text{C}_2\text{H}_2(g) + 2 \text{H}_2(g) \rightarrow \text{C}_2\text{H}_6(g)$
(E) $\text{MgSO}_3(s) \rightarrow \text{MgO}(s) + \text{SO}_2(g)$ ✓

42. Of the following reactions, which involves the largest decrease in entropy?

- (A) $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$
(B) $2 \text{CO}(g) + \text{O}_2(g) \rightarrow 2 \text{CO}_2(g)$
(C) $\text{Pb}(\text{NO}_3)_2(s) + 2 \text{KI}(s) \rightarrow \text{PbI}_2(s) + 2 \text{KNO}_3(s)$
(D) $\text{C}_3\text{H}_8(g) + 5 \text{O}_2(g) \rightarrow 3 \text{CO}_2(g) + 4 \text{H}_2\text{O}(g)$
(E) $4 \text{La}(s) + 3 \text{O}_2(g) \rightarrow 2 \text{La}_2\text{O}_3(s)$ ✓

43. $\text{C}(\text{diamond}) \rightarrow \text{C}(\text{graphite}) \quad \Delta G^\circ = -2.9 \text{ kJ/mol}_{rxn}$

Which of the following best explains why the reaction represented above is not observed to occur at room temperature?

Chapter 08 Part 3: Entropy and Gibbs

- (A) The rate of the reaction is extremely slow because of the relatively small value of ΔG° for the reaction.
- (B) The entropy of the system decreases because the carbon atoms in graphite are less ordered than those in diamond.
- (C) The reaction has an extremely large activation energy due to strong three-dimensional bonding among carbon atoms in diamond. ✓
- (D) The reaction does not occur because it is not thermodynamically favorable.
44. As a sample of $\text{KNO}_3(s)$ is stirred into water at 25°C , the compound dissolves endothermically. Which of the following best helps to explain why the process is thermodynamically favorable at 25°C ?
- (A) All endothermic processes are thermodynamically favorable.
- (B) Stirring the solution during dissolution adds the energy needed to drive an endothermic process.
- (C) Dissolving the salt decreases the enthalpy of the system.
- (D) Dissolving the salt increases the entropy of the system. ✓
45.
$$2 \text{H}_2(g) + \text{O}_2(g) \rightarrow 2 \text{H}_2\text{O}(g)$$
- When $\text{H}_2(g)$ and $\text{O}_2(g)$ are mixed together in a rigid reaction vessel at 25°C , no reaction occurs. When the mixture is sparked, however, the gases react vigorously according to the equation above, releasing heat. Which of the following statements correctly explains why the spark is needed for the reaction to occur when the gases are originally at 25°C ?
- (A) The reaction is not thermodynamically favorable at 25°C .
- (B) ΔH° for the reaction has a large positive value at 25°C .
- (C) ΔS° for the reaction has a large negative value at 25°C .
- (D) The reaction has a large activation energy at 25°C . ✓

Chapter 08 Part 3: Entropy and Gibbs



A student investigates the reaction between $\text{Ag}(s)$ and $\text{HNO}_3(aq)$ represented by the equation above.

- Predict the sign of the entropy change, ΔS° , for the reaction. Justify your answer.
- Use the information in the table below to calculate the value ΔH_{rxn}° , the standard enthalpy change for the reaction, in kJ/mol_{rxn} .

Substance	$\text{HNO}_3(aq)$	$\text{AgNO}_3(aq)$	$\text{NO}(g)$	$\text{H}_2\text{O}(l)$
Standard Heat of Formation, ΔH_f° (kJ/mol)	-207	-101	90.	-286

- Based on your answers to parts (a) and (b), is the reaction more likely to be thermodynamically favorable at 25°C , or at 95°C ? Justify your answer.
- The student runs the reaction using a 3 to 4 mole ratio of $\text{Ag}(s)$ to $\text{HNO}_3(aq)$. Suggest a method the student can use to isolate solid AgNO_3 from the other products of the reaction.

Part A

1 point is earned for the correct answer and a valid justification.

The entropy change is positive because the reaction has one mole of gas in the products and none in the reactants.



0	1
---	---

Student earns 1/1 point(s)

1 point is earned for the correct answer and a valid justification.

The entropy change is positive because the reaction has one mole of gas in the products and none in the reactants.

Part B

1 point is earned for the correct calculation of the value of ΔH° .

$$\begin{aligned}\Delta H^\circ &= 3(-101) + 90. + 2(-286) - 4(-207) \\ &= 43 \text{ kJ/mol}_{rxn}\end{aligned}$$

Chapter 08 Part 3: Entropy and Gibbs



0	1
---	---

Student earns 1/1 point(s)

1 point is earned for the correct calculation of the value of ΔH° .

$$\begin{aligned}\Delta H^\circ &= 3(-101) + 90. + 2(-286) - 4(-207) \\ &= 43 \text{ kJ/molrxn}\end{aligned}$$

Part C

1 point is earned for the correct answer and a valid justification.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

The reaction is more likely to be favorable at 95°C. At the higher temperature, the term $T\Delta S^\circ$ is larger and positive; thus, when subtracted from ΔH° , the value of ΔG° is more likely to be negative.



0	1
---	---

Student earns 1/1 point(s)

1 point is earned for the correct answer and a valid justification.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

The reaction is more likely to be favorable at 95°C. At the higher temperature, the term $T\Delta S^\circ$ is larger and positive; thus, when subtracted from ΔH° , the value of ΔG° is more likely to be negative.

Part D

1 point is earned for a valid method.

The student can evaporate the water, leaving behind solid silver nitrate.



0	1
---	---

Student earns 1/1 point(s)

Chapter 08 Part 3: Entropy and Gibbs

1 point is earned for a valid method.

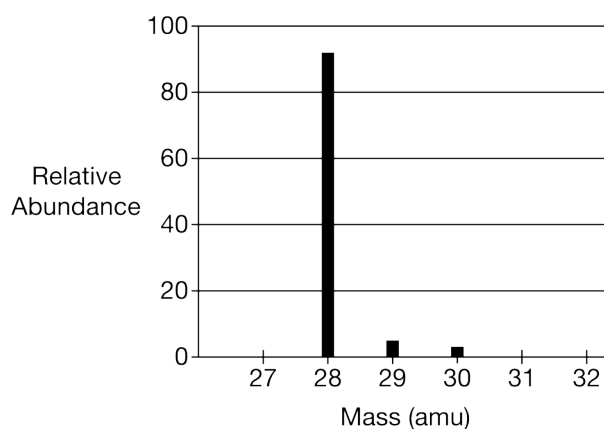
The student can evaporate the water, leaving behind solid silver nitrate.

Chapter 08 Part 3: Entropy and Gibbs

47. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about the element Si and some of its compounds.

- (a) The mass spectrum of a pure sample of Si is shown below.



- (i) How many protons and how many neutrons are in the nucleus of an atom of the most abundant isotope of Si?

- (ii) Write the ground-state electron configuration of Si.

Two compounds that contain Si are SiO_2 and SiH_4 .

- (b) At 161 K, SiH_4 boils but SiO_2 remains as a solid. Using principles of interparticle forces, explain the difference in boiling points.

At high temperatures, SiH_4 decomposes to form solid silicon and hydrogen gas.

- (c) Write a balanced equation for the reaction.

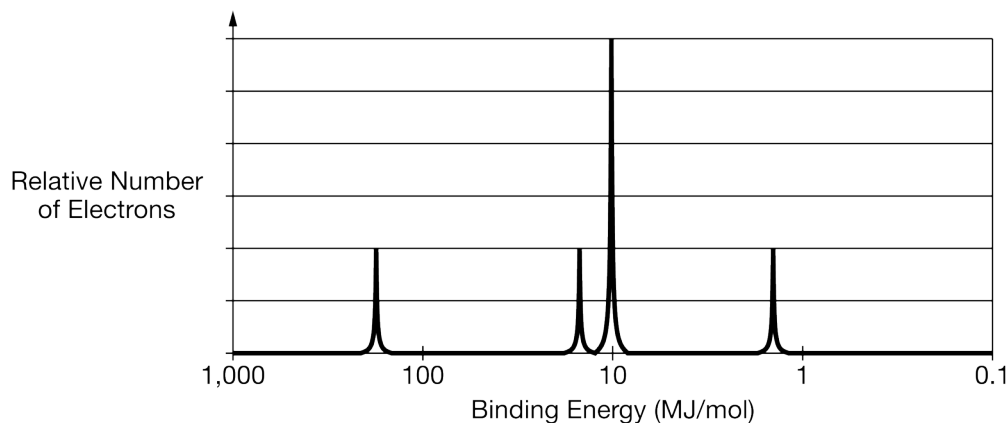
A table of absolute entropies of some substances is given below.

Substance	S° (J/(mol · K))
$\text{H}_2(g)$	131
$\text{Si}(s)$	18
$\text{SiH}_4(g)$	205

- (d) Explain why the absolute molar entropy of $\text{Si}(s)$ is less than that of $\text{H}_2(g)$.

Chapter 08 Part 3: Entropy and Gibbs

- (e) Calculate the value, in $\text{J}/(\text{mol} \cdot \text{K})$, of ΔS° for the reaction.
- (f) The reaction is thermodynamically favorable at all temperatures. Explain why the reaction occurs only at high temperatures.
- (g) A partial photoelectron spectrum of pure Si is shown below. On the spectrum, draw the missing peak that corresponds to the electrons in the $3p$ sublevel.



- (h) Using principles of atomic structure, explain why the first ionization energy of Ge is lower than that of Si.
- (i) A single photon with a wavelength of $4.00 \times 10^{-7} \text{ m}$ is absorbed by the Si sample. Calculate the energy of the photon in joules.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **all** the following criteria.

- ☐ The response provides a correct answer.
· 14 protons and 14 neutrons
- ☐ The response provides a correct answer.
· $1s^2 2s^2 2p^6 3s^2 3p^2$
OR
· $[\text{Ne}] 3s^2 3p^2$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 08 Part 3: Entropy and Gibbs

0	1
---	---

The response provides a correct explanation.

· SiH_4 is composed of molecules, for which the only intermolecular forces are London dispersion forces. SiO_2 is a network covalent compound with covalent bonds between silicon and oxygen atoms. London dispersion forces are much weaker than covalent bonds so SiH_4 boils at a much lower temperature than SiO_2 .

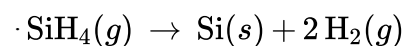
Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct balanced equation (state symbols not required).

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

· The $\text{H}_2(g)$ molecules are more highly dispersed than the $\text{Si}(s)$ atoms and, therefore, have a higher absolute molar entropy. Silicon is a solid; therefore, its atoms are in fixed positions, are less dispersed, and have a lower absolute molar entropy.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

Chapter 08 Part 3: Entropy and Gibbs

$$\Delta S_{\text{rxn}}^{\circ} = (18 + 2(131)) - 205 = +75 \text{ J}/(\text{mol}_{\text{rxn}} \cdot \text{K})$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

· High temperature is required for the reactant particles to have sufficient thermal energy to overcome the activation energy of the reaction.

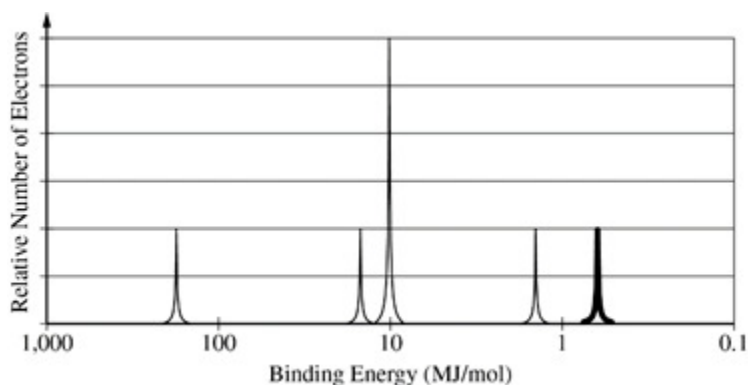
Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct peak height and location. (The peak should be drawn to the right of the other peaks, and it should reach the second line above the horizontal axis.)

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

Chapter 08 Part 3: Entropy and Gibbs

The response provides a correct explanation.

· The valence electrons of a Ge atom occupy a higher shell ($n = 4$) than those of a Si atom ($n = 3$), so the average distance between the nucleus and the valence electrons is greater in Ge than in Si. This greater separation results in weaker Coulombic attractions between the Ge nucleus and its valence electrons, making them less tightly bound and, therefore, easier to remove compared to those in Si.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct calculated value.

$$\cdot E = h\nu = h\left(\frac{c}{\lambda}\right) = 6.626 \times 10^{-34} \text{ J} \cdot \text{s} \left(\frac{2.998 \times 10^8 \text{ m s}^{-1}}{4.00 \times 10^{-7} \text{ m}}\right) = 4.97 \times 10^{-19} \text{ J}$$

Chapter 08 Part 3: Entropy and Gibbs

48. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

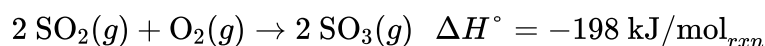
The following questions relate to sulfur and some of its compounds.

- (a) Write the balanced equation for the combustion of $\text{S}_8(s)$ to form $\text{SO}_2(g)$.
- (b) Calculate the volume of $\text{O}_2(g)$, measured at 1.00 atm and 298 K, that is required to completely combust a 500.0 g sample of pure $\text{S}_8(s)$.
- (c) A student claims that the combustion of S_8 is an oxidation-reduction reaction. Justify the claim by identifying the oxidation numbers of sulfur and oxygen both before and after the reaction.
- (d) In the box below, draw a Lewis electron-dot diagram for one valid resonance structure of SO_2 .

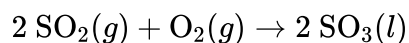


- (e) Based on the diagram you drew in part (d), what is the approximate oxygen-sulfur-oxygen bond angle in SO_2 ?

SO_2 can be oxidized to form SO_3 according to the following equation.



- (f) Is the value of ΔS° for the reaction represented above positive or negative? Justify your answer.



Chapter 08 Part 3: Entropy and Gibbs

(g) Is the magnitude of ΔH° for the reaction to form $\text{SO}_3(l)$, represented above, greater than, less than, or equal to the magnitude of ΔH° for the reaction to form $\text{SO}_3(g)$? Justify your answer.

(h) Based on the information above, how does the thermodynamic favorability of the reaction change as the temperature of the reaction system is decreased? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response contains the equation $\text{S}_8(s) + 8 \text{O}_2(g) \rightarrow 8 \text{SO}_2(g)$.
(Note: phase designations are not required.)

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the following criteria.

- ☐ There is a set up and calculation of the number of moles of $\text{O}_2(g)$ required as follows.

$$500.0 \text{ g S}_8 \times \frac{1 \text{ mol S}_8}{256.5 \text{ g S}_8} \times \frac{8 \text{ mol O}_2}{1 \text{ mol S}_8} = 15.60 \text{ mol O}_2$$
- ☐ There is a set up and calculation of the volume of O_2 as follows.

$$V = \frac{nRT}{P} = \frac{(15.60 \text{ mol})(0.08206 \text{ L atm mol}^{-1} \text{ K})(298 \text{ K})}{1.00 \text{ atm}} = 381 \text{ L}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes all four oxidation numbers

· Before the reaction, $\text{S} = 0$, $\text{O} = 0$

Chapter 08 Part 3: Entropy and Gibbs

· After the reaction, $S = +4$, $O = -2$

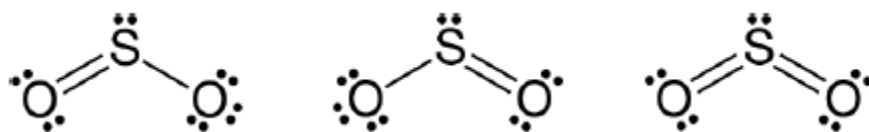
Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes any one of the following three diagrams.

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets either of the following two criteria.

- ☐ The response indicates that the bond angle is 120°
- ☐ The response indicates an angle that is consistent with the incorrect diagram drawn in part (d).

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the value of ΔS° is negative because the reactants are three moles of gas but the products are only two moles of gas.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 08 Part 3: Entropy and Gibbs



0	1
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The response indicates that the magnitude of ΔH° for the production of $\text{SO}_3(l)$ is greater because the enthalpy of $\text{SO}_3(l)$ is lower than the enthalpy of $\text{SO}_3(g)$ (by an amount equal to the enthalpy of vaporization of $\text{SO}_3(l)$), which makes the difference between the enthalpy of the products and the enthalpy of the reactants a larger amount.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



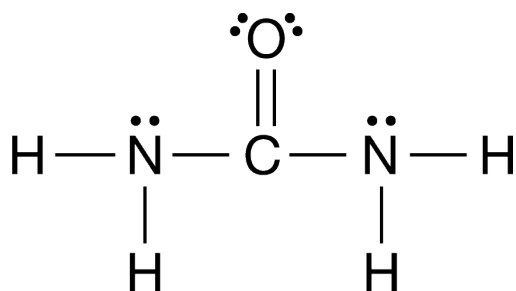
0	1
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The response meets both of the following criteria.

- ☐ The response indicates that the thermodynamic favorability of the reaction increases with decreasing temperature.
- ☐ The response includes a valid justification such as the example below.
 $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$. Assume that the values of both ΔH° and ΔS° are constant. As the value of T decreases, so does the value of the term $T\Delta S^\circ$, making the quantity $(\Delta H^\circ - T\Delta S^\circ)$ more negative, thereby causing the thermodynamic favorability to increase.

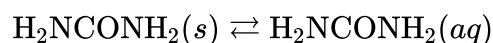
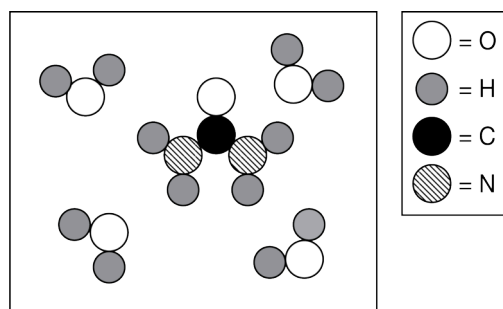
Chapter 08 Part 3: Entropy and Gibbs

49. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The compound urea, H_2NCONH_2 , is widely used in chemical fertilizers. The complete Lewis electron-dot diagram for the urea molecule is shown above.

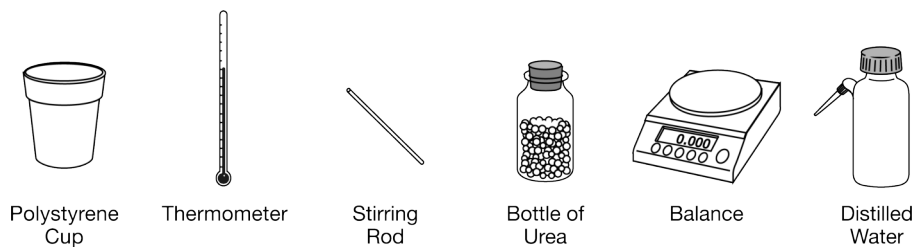
- (a) Identify the hybridization of the valence orbitals of the carbon atom in the urea molecule.
- (b) Urea has a high solubility in water, due in part to its ability to form hydrogen bonds. A urea molecule and four water molecules are represented in the box below. Draw ONE dashed line (----) to indicate a possible location of a hydrogen bond between a water molecule and the urea molecule.



The dissolution of urea is represented by the equation above. A student determines that 5.39 grams of H_2NCONH_2 (molar mass 60.06 g/mol) can dissolve in water to make 5.00 mL of a saturated solution at $20.^\circ\text{C}$.

- (c) Calculate the concentration of urea, in mol/L, in the saturated solution at $20.^\circ\text{C}$.
- (d) The student also determines that the concentration of urea in a saturated solution at 25°C is 19.8 M. Based on this information, is the dissolution of urea endothermic or exothermic? Justify your answer in terms of Le Chatelier's principle.

Chapter 08 Part 3: Entropy and Gibbs



(e) The equipment shown above is provided so that the student can determine the value of the molar heat of solution for urea. Knowing that the specific heat of the solution is $4.18 \text{ J}/(\text{g} \cdot ^\circ\text{C})$, list the specific measurements that are required to be made during the experiment.

	$S^\circ \text{ (J}/(\text{mol} \cdot \text{K}))$
$\text{H}_2\text{NCONH}_2(s)$	104.6
$\text{H}_2\text{NCONH}_2(aq)$	?

(f) The entropy change for the dissolution of urea, $\Delta S_{\text{soln}}^\circ$, is $70.1 \text{ J}/(\text{mol} \cdot \text{K})$ at 25°C . Using the information in the table above, calculate the absolute molar entropy, S° , of aqueous urea.

(g) Using particle-level reasoning, explain why $\Delta S_{\text{soln}}^\circ$ is positive for the dissolution of urea in water.

(h) The student claims that ΔS° for the process contributes to the thermodynamic favorability of the dissolution of urea at 25°C . Use the thermodynamic information above to support the student's claim.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates sp^2 .

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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There is a dashed line connecting a hydrogen atom in water to a nitrogen or oxygen atom in urea or a dashed line connecting an oxygen atom in water to a hydrogen atom in urea.

Chapter 08 Part 3: Entropy and Gibbs

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the following criteria.

- ☐ The response shows the following calculation.
$$5.39 \text{ g H}_2\text{NCONH}_2 \times \frac{1 \text{ mol}}{60.06 \text{ g}} = 0.0897 \text{ mol}$$
- ☐ The response shows the following calculation.
$$\frac{0.0897 \text{ mol}}{0.00500 \text{ L}} = 17.9 \text{ M}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ The response indicates that the dissolution is endothermic.
- ☐ The response gives an explanation such as the following. If a saturated solution of urea is heated, then the equilibrium system is stressed. The stress is counteracted by the endothermic dissolution of more urea.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the following criteria.

- ☐ The response indicates that the mass of urea and the mass of water are required.
- ☐ The response indicates that the initial temperature of the water and the final temperature of the solution are required.

Chapter 08 Part 3: Entropy and Gibbs**Part F**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the following calculations.

$$\Delta S_{\text{soln}}^{\circ} = S^{\circ}(\text{H}_2\text{NCONH}_2(aq)) - S^{\circ}(\text{H}_2\text{NCONH}_2(s))$$

$$70.1 \text{ J}/(\text{mol} \cdot \text{K}) = S^{\circ}(\text{H}_2\text{NCONH}_2(aq)) - 104.6 \text{ J}/(\text{mol} \cdot \text{K})$$

$$S^{\circ}(\text{H}_2\text{NCONH}_2(aq)) = 174.7 \text{ J}/(\text{mol} \cdot \text{K})$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that urea molecules in solution have a greater number of possible arrangements than in solid urea, and that this increased number of arrangements corresponds to a positive $\Delta S_{\text{soln}}^{\circ}$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives an explanation such as the following.

Thermodynamic favorability for a process at standard conditions is determined by the sign of ΔG° , with $\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$. Since ΔS° is positive, the $T\Delta S^{\circ}$ term makes the value of ΔG° smaller and thus makes the dissolution more thermodynamically favorable.